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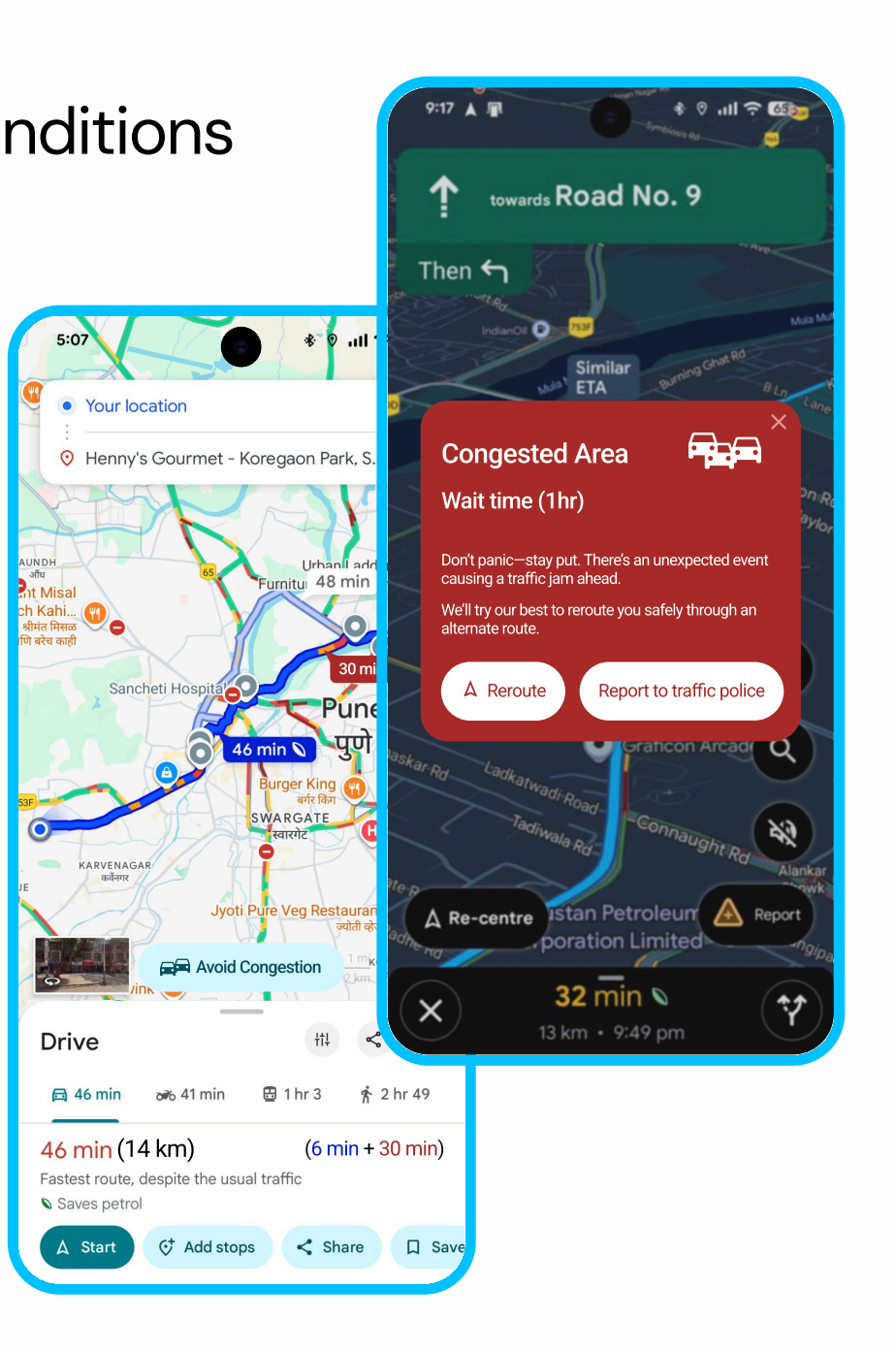
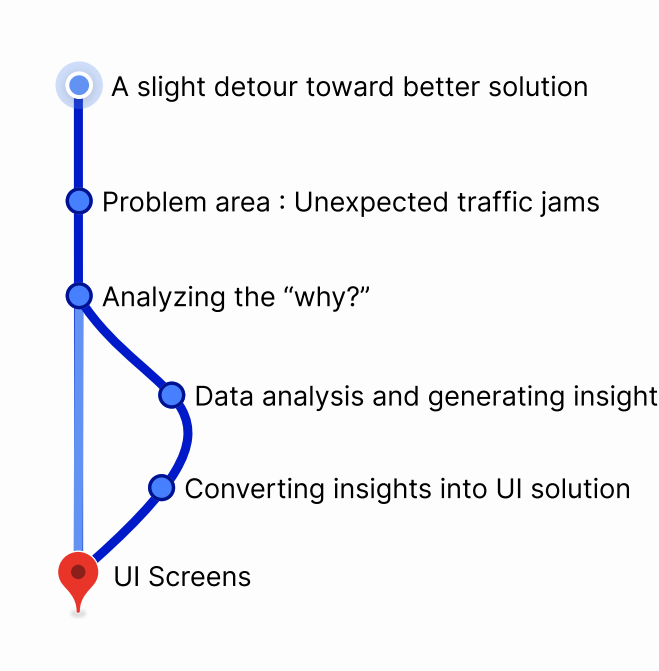
Information and Data Viz Designer

Suchālak

Enhancing *Google Maps' accuracy* for Indian traffic conditions by integrating *emotional cues* into navigation insights.


towards Road No. 9
Then
Congested Area
Wait time (1hr)
Reroute
Report to traffic police
Re-centre
Report
32 min
Drive
46 min (14 km)
Start
Add stops
Share
Save


A slight detour toward better solution
Problem area : Unexpected traffic jams
Analyzing the "why?"
Data analysis and generating insights
Converting insights into UI solution
UI Screens

Tools used

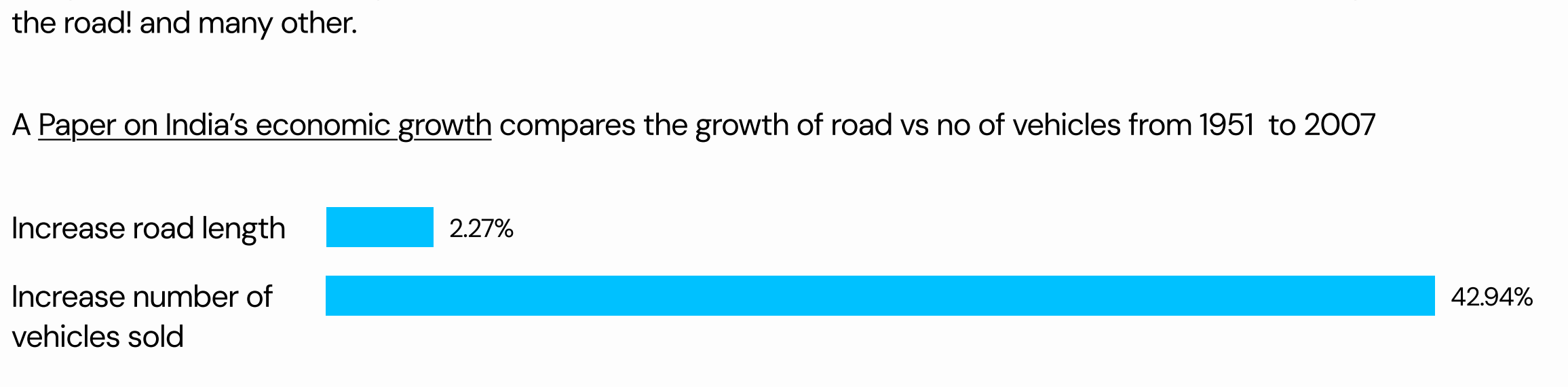
Figma Tableau Python Power BI Google forms

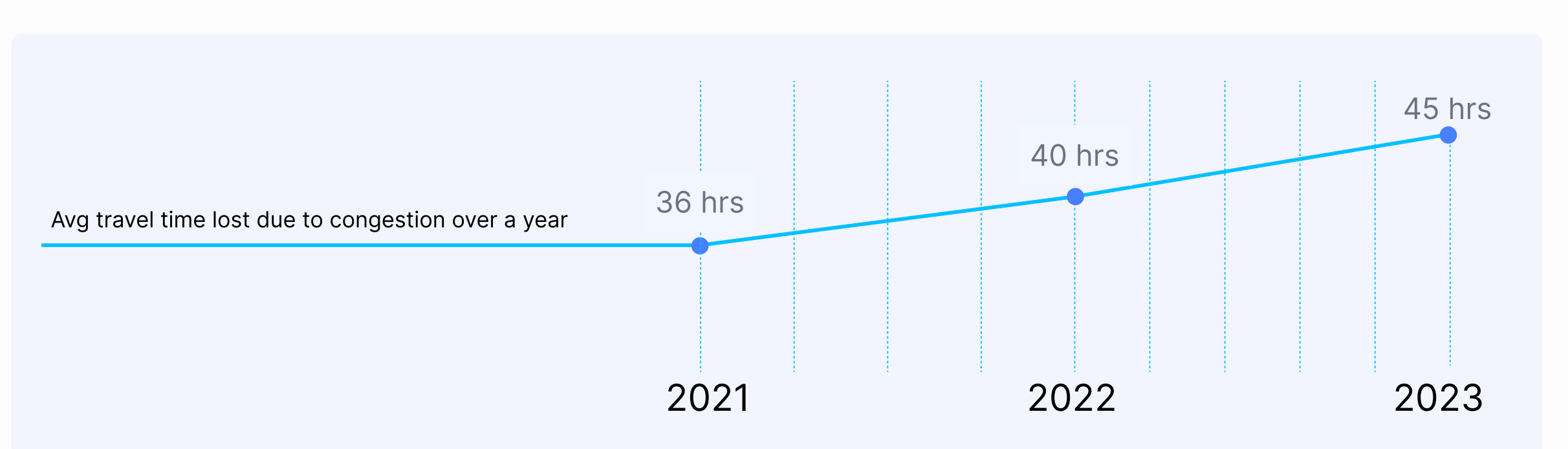
#UI Design #Data-driven Design #Behavioral UX

01 Problem Area

In the Indian context, real-time traffic estimates frequently become inaccurate because of sudden and unpredictable congestion. And these congestions happens due to multiple reasons- festivals, accidents, random light pole fell and blocked the road! and many other.

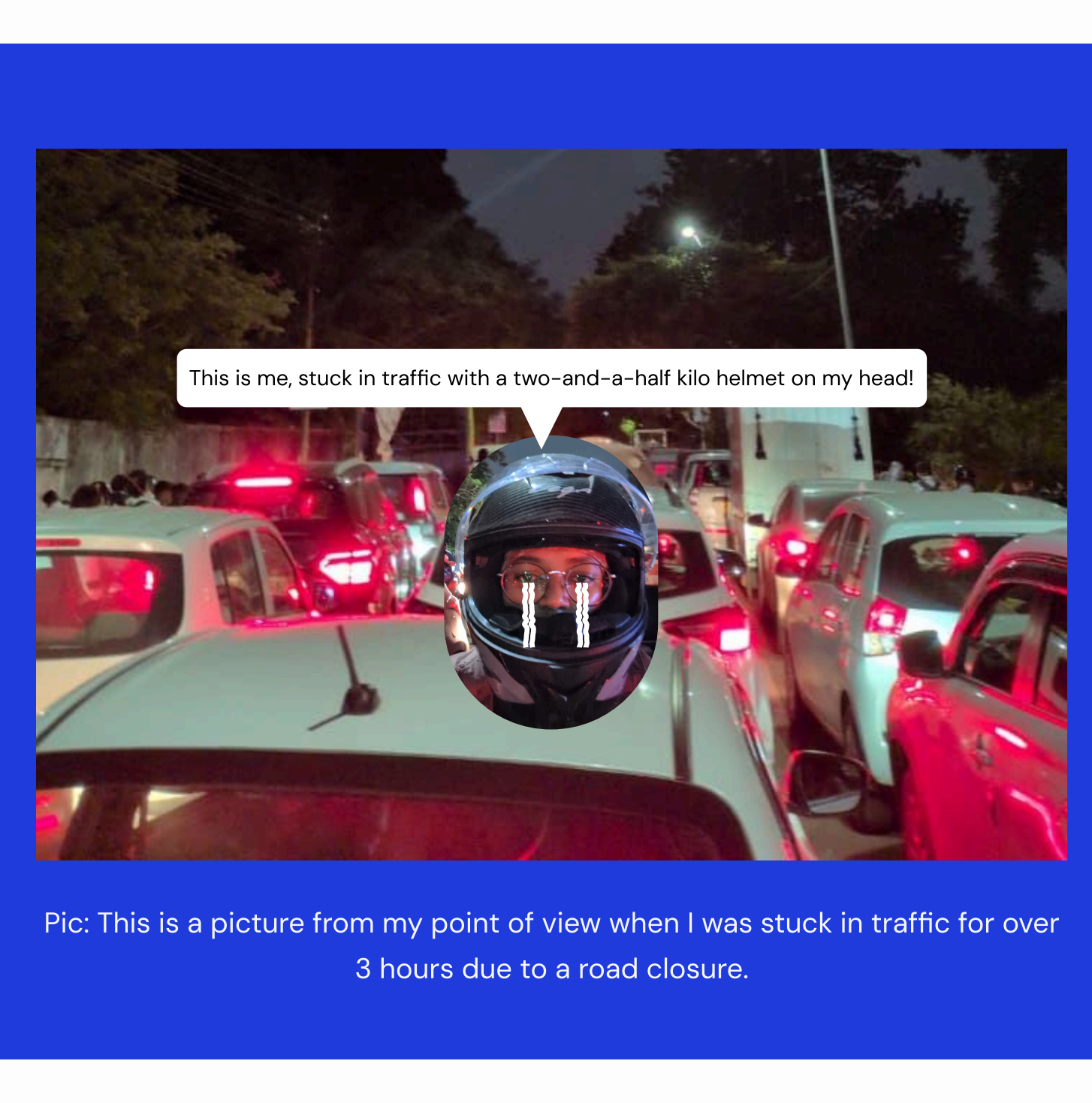
A Paper on India's economic growth compares the growth of road vs no of vehicles from 1951 to 2007





On average 4hrs increasing year by year per driver

This is not just me it's you, it's all of us sitting in the same traffic, feeling the same frustration, watching the minutes slip away in long congestions.


This is me, stuck in traffic with a two-and-a-half kilo helmet on my head!

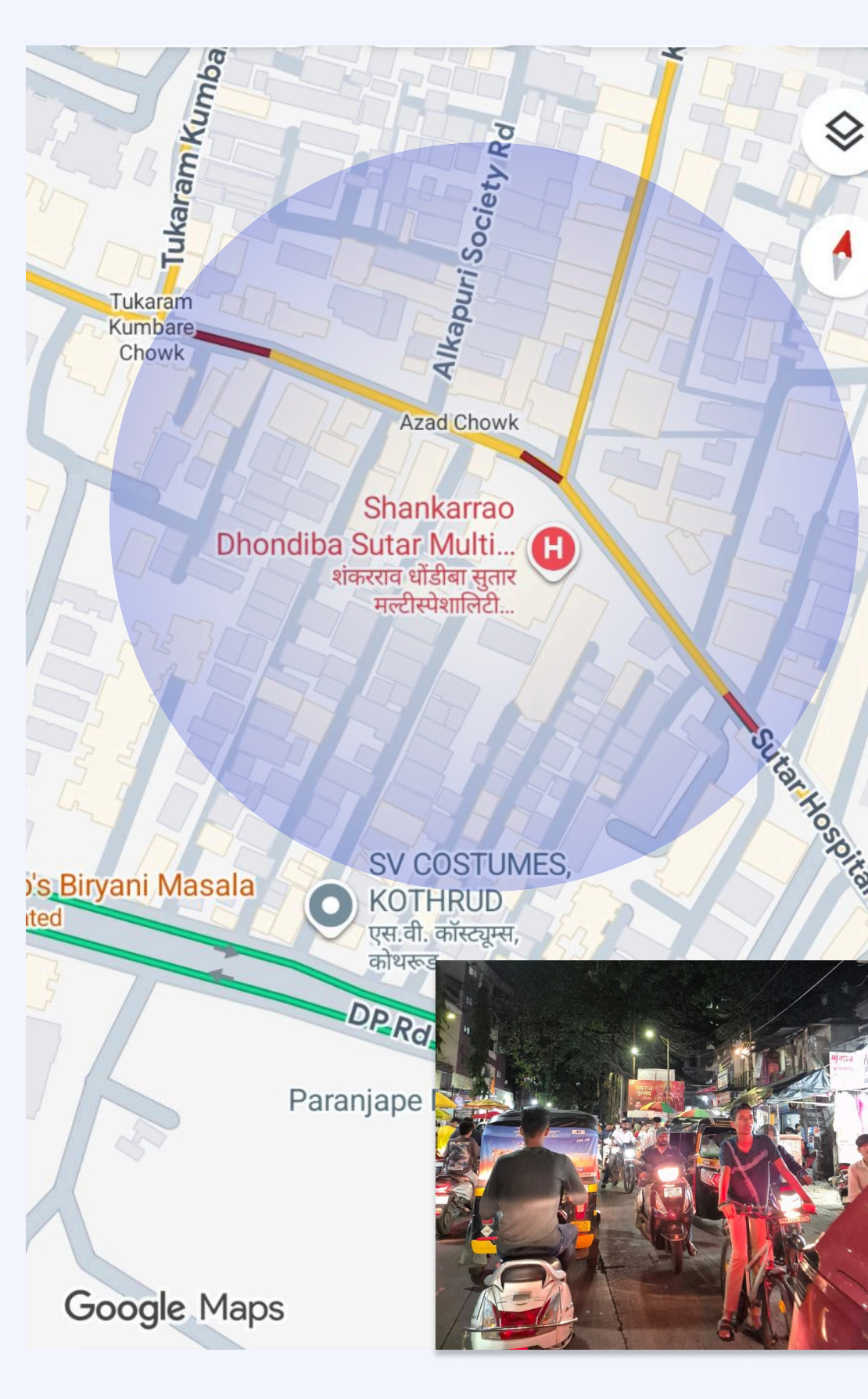
Pic: This is a picture from my point of view when I was stuck in traffic for over 3 hours due to a road closure.

02 Identifying the "Gap"

While understanding the underlying problem, I identified a gap between what we experience on the road and what Google Maps tells us. Taking the congestion issues in my own area as an example, I can clearly pinpoint this gap.

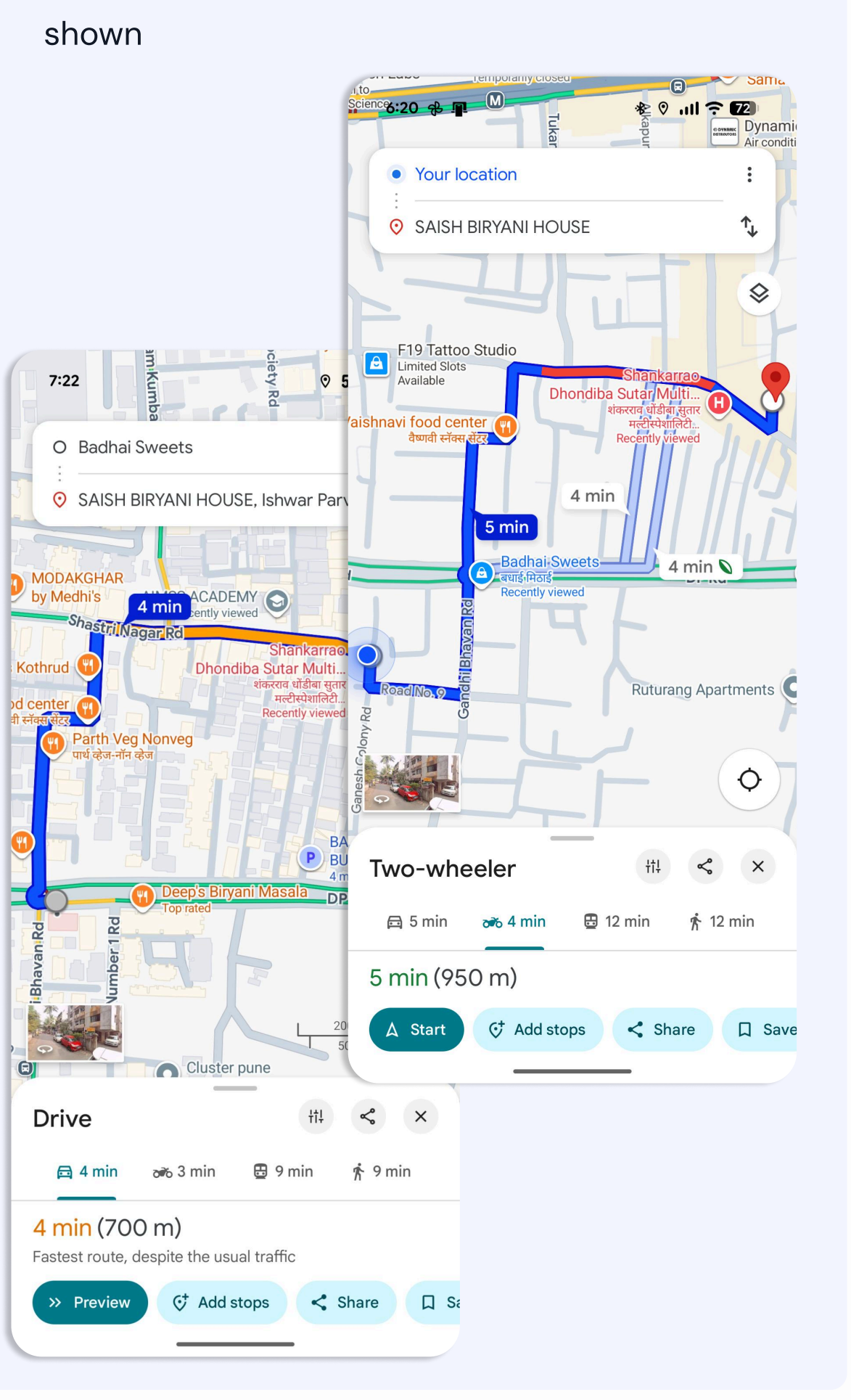
Real Persona

- Avoiding going thorough this area
- Knowing that once stuck it take at least **20 min** to get out from traffic jam


Tukaram Kumbar
Tukaram Kumbare Chowk
Alkapuri Society Rd
Azad Chowk
Shankarrao Dhondiba Sutar Multi...
Sutar Hospital Rd
SV COSTUMES KOTHRUD
DP Rd
Paranjape
Google Maps

Google's persona

- Shows that it take 4-5 min depending on vehicle
- Estimates time doesn't show the possibility of getting stuck
- If you're stuck in traffic, the exact wait time isn't shown


Your location
SAISH BIRYANI HOUSE
Badhai Sweets
5 min
4 min
Two-wheeler
5 min (950 m)
Start
Add stops
Share
Save
Drive
4 min (700 m)
Fastest route, despite the usual traffic
Preview

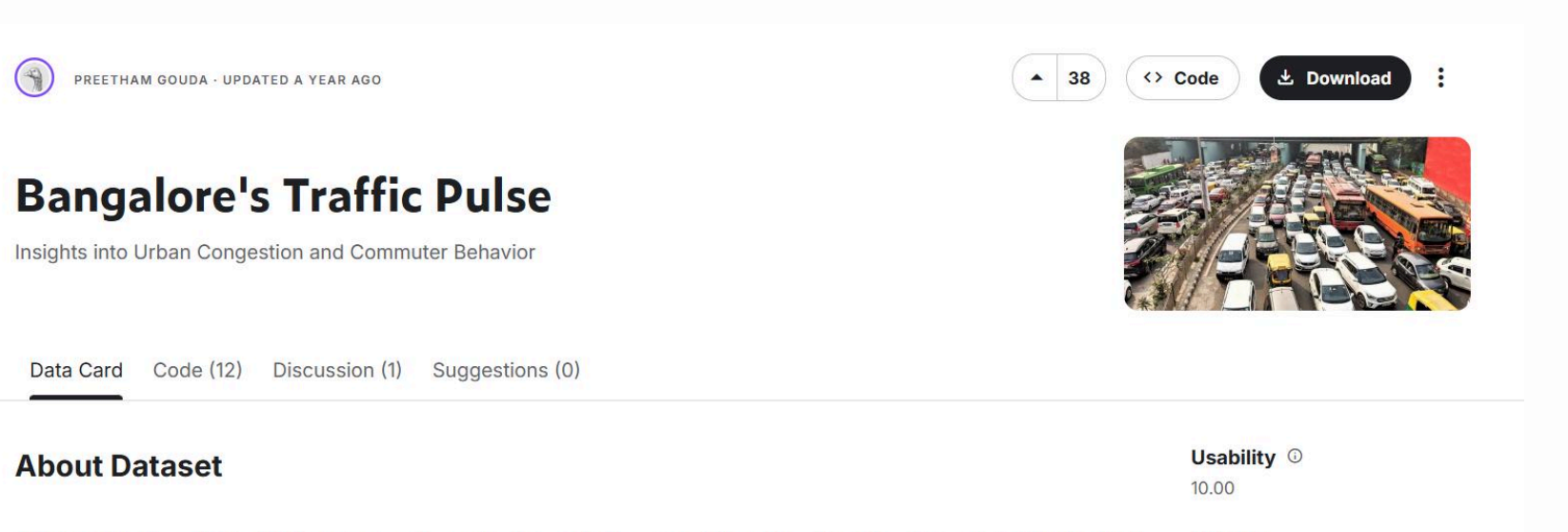
Experience ——— Gap ——— Showcase

03 Why congestion happens?

To understand this particular problem we need to find pattern *where* and *how* the congestion happens in a particular city. Which can be found by data analysis of traffic congestion data sets.

From the data analysis it was evident that the amount of traffic is 100 times more than the capacity of the roads

For detailed data analysis click here



Bangalore's Traffic Pulse

Insights into Urban Congestion and Commuter Behavior

About Dataset

This dataset captures detailed traffic patterns across major roads and intersections in Bangalore. It includes metrics such as traffic volume, speed, congestion levels, and environmental impact, providing a comprehensive view of the city's transportation dynamics. Ideal for urban planners and smart city applications.

Usability 10.00

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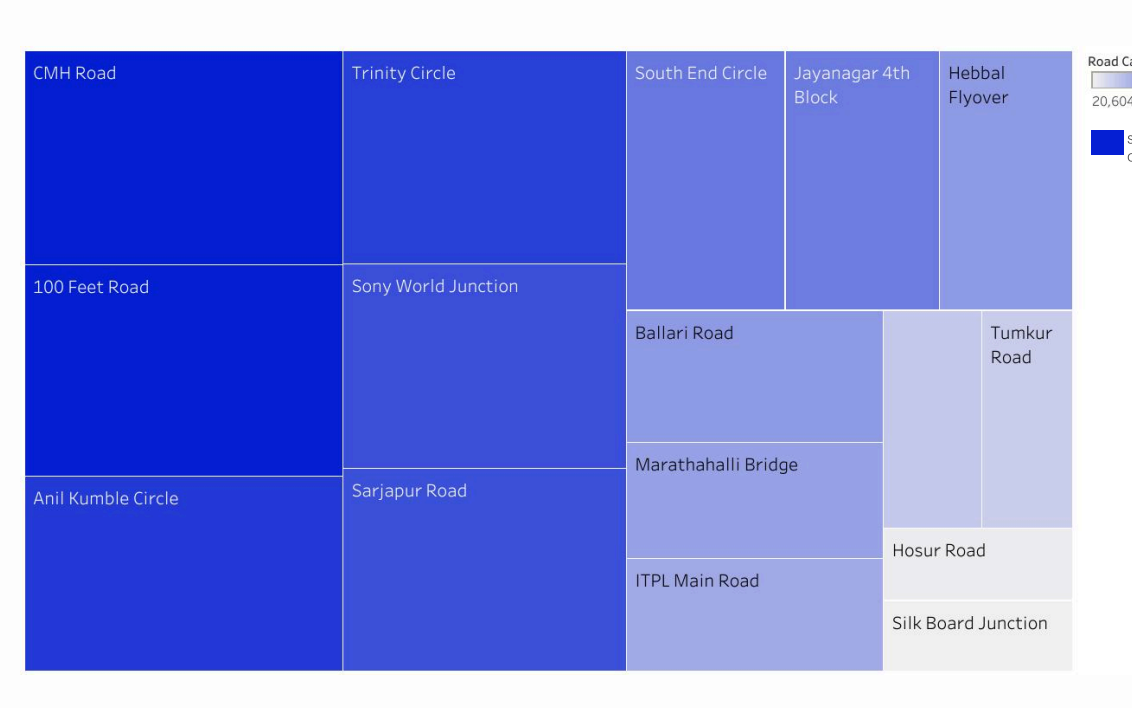
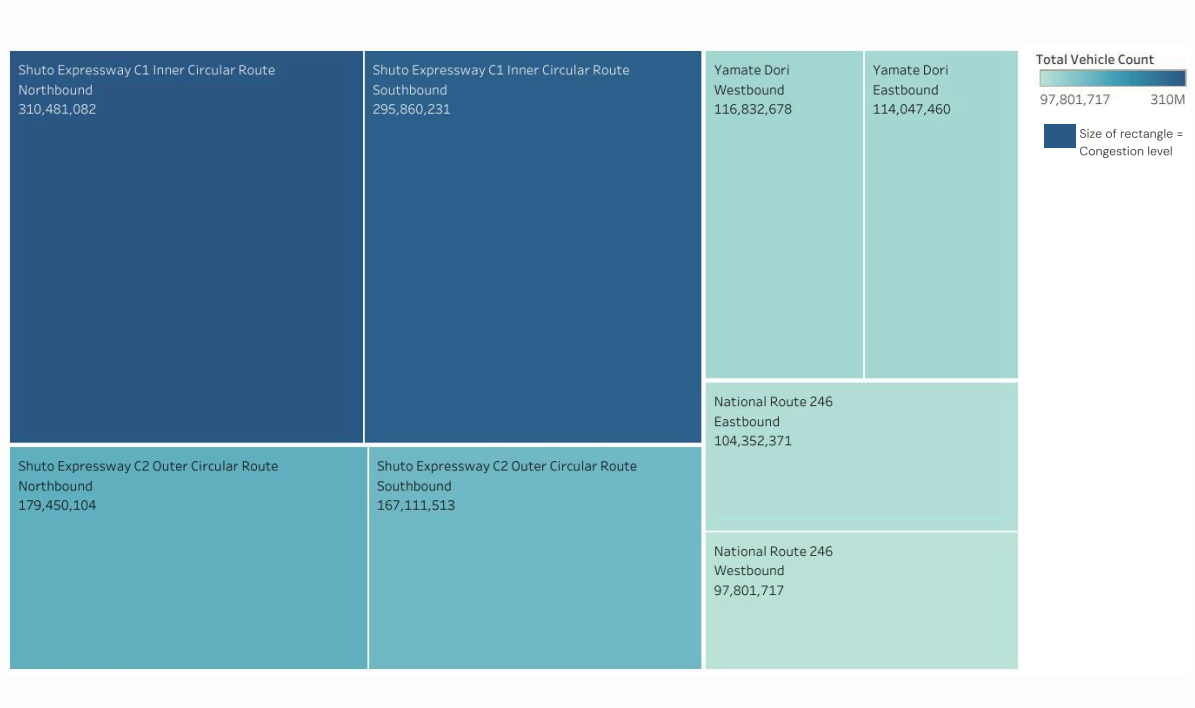
Expected update frequency Never

38
Code
Download
Data Card
Code (13)
Discussion (1)
Suggestions (0)

Below is tree map shows the area wise traffic distribution at peak travel time in Bangalore and Tokyo



Bangalore



Tokyo

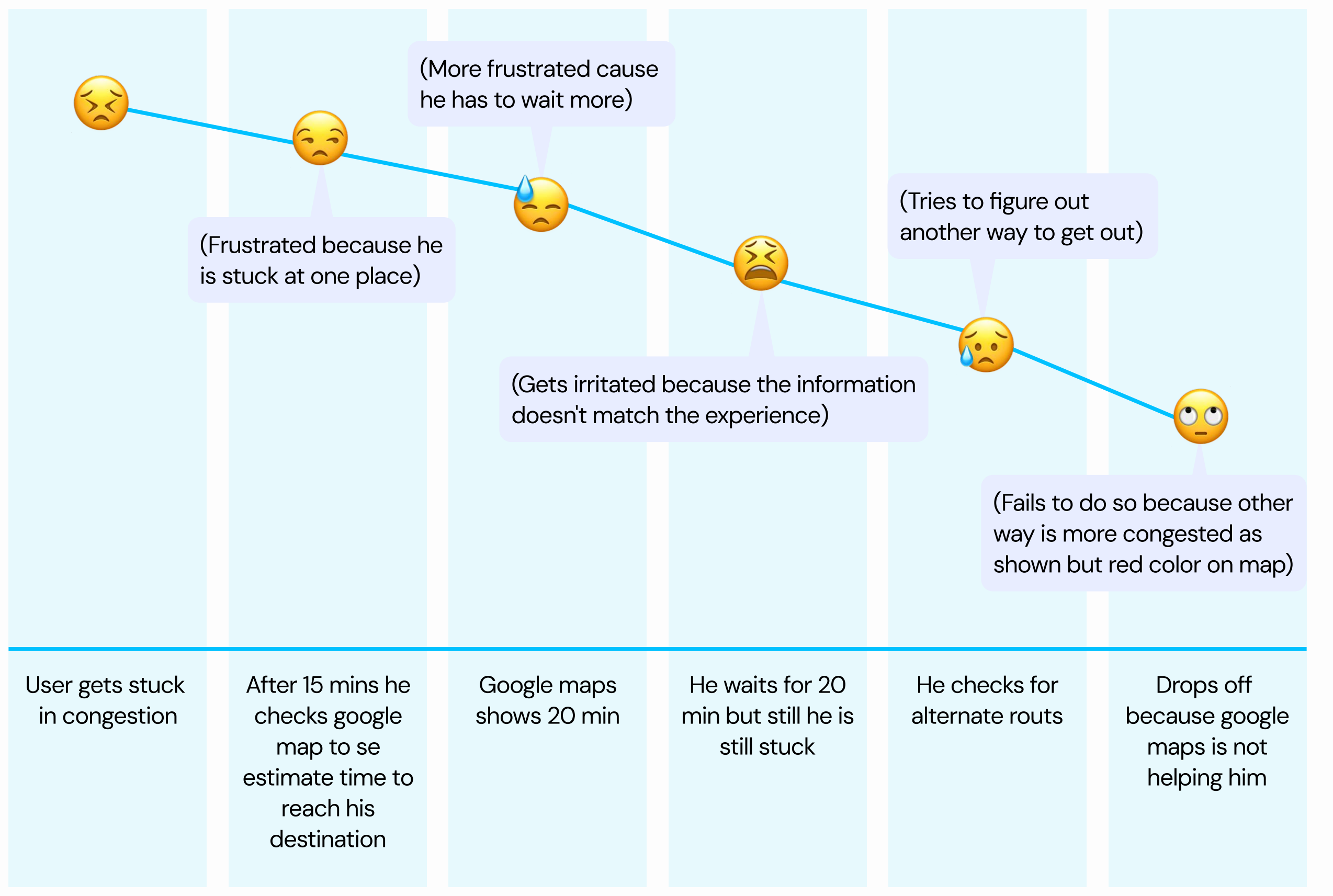
Few points to consider form this comparison:
1. Tokyo has 50% traffic more population than Bangalore
2. Still Tokyo manages to keep the congestion intensity dispersed with area in consideration
3. Bangalore gets congestion at all areas at the same time

> " This comparison shows that Tokyo's traffic density data is accurately captured which is somewhat failed to do in Bangalore(and other metropolitan cities in India)"

04 Understanding the people

To understand why the data is not captured, I conducted a survey on how people navigate a congestion and how they use google map in that scenario. Also understanding the emotional range of user as to how they perceive any information during that period of time

This can be better explained with the help of *Customer Journey Map* effectively.


(Frustrated because he is stuck at one place)
(More frustrated cause he has to wait more)
(Gets irritated because the information doesn't match the experience)
(Tries to figure out another way to get out)
(Fails to do so because other way is more congested as shown but red color on map)
User gets stuck in congestion
After 15 mins he checks google map to see estimate time to reach his destination
Google maps shows 20 min
He waits for 20 min but still he is still stuck
He checks for alternate routs
Drops off because google maps is not helping him

This is cumulative insight for detailed analysis click here

High cognitive load causes user frustration and drop-offs, leading to missing data points and reducing the overall accuracy of traffic data collection. Which works against google maps traffic prediction in Indian cities

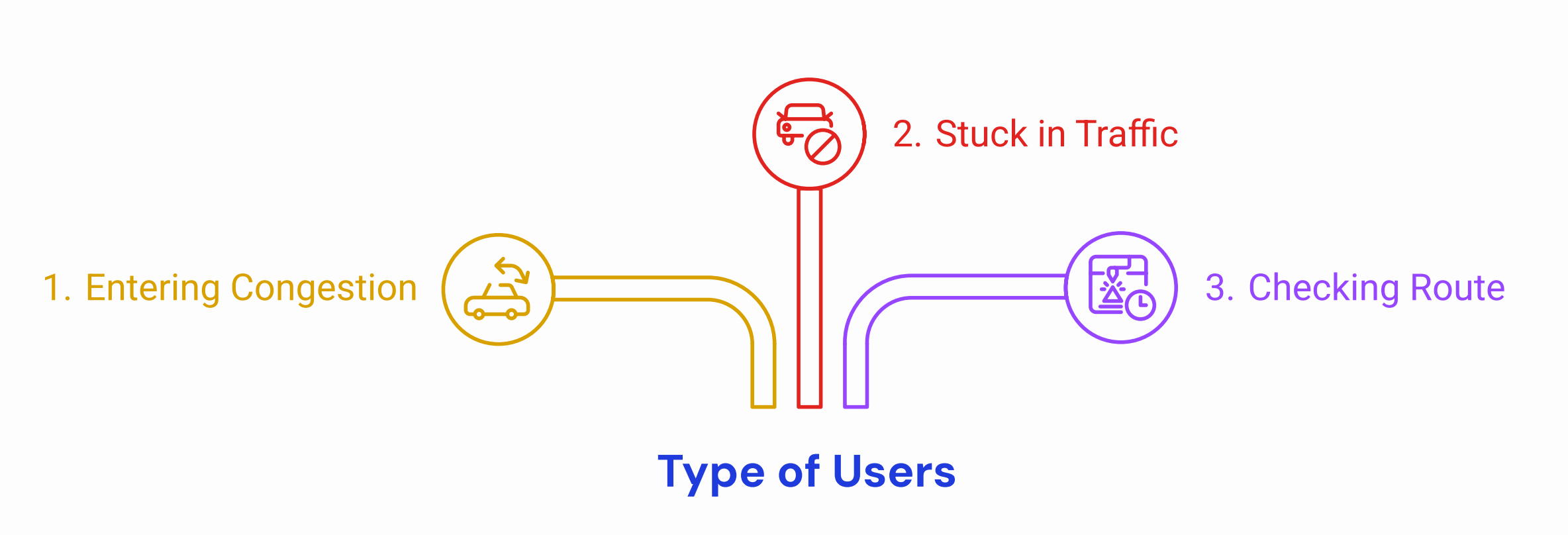
05 Making Google maps a "Suchālak"

Suchālak means an efficient guide. However, effective guidance is not just about directions, it also requires sensitivity to users' emotions and real-time traffic stress. At present, Google Maps falls short in providing the contextual and emotional cues required for navigating complex Indian traffic scenarios.

So depending on the usages of google maps there are three types of users which primarily comes from different scenarios in which they use google maps.

🤓 Categorization of solution based on type of user

Required information cues to support decision-making :
- Emotional Assurance
- Giving exact "**Time spent in congestion**" as a feature


1. Entering Congestion
2. Stuck in Traffic
3. Checking Route
Type of Users

Flow 1 needs :
- Alert based on travel history
- Identifiable sound pop-up notification

Flow 2 needs :
- Show duration of congestion
- Contact traffic authorities
- Show re-route through without congestion

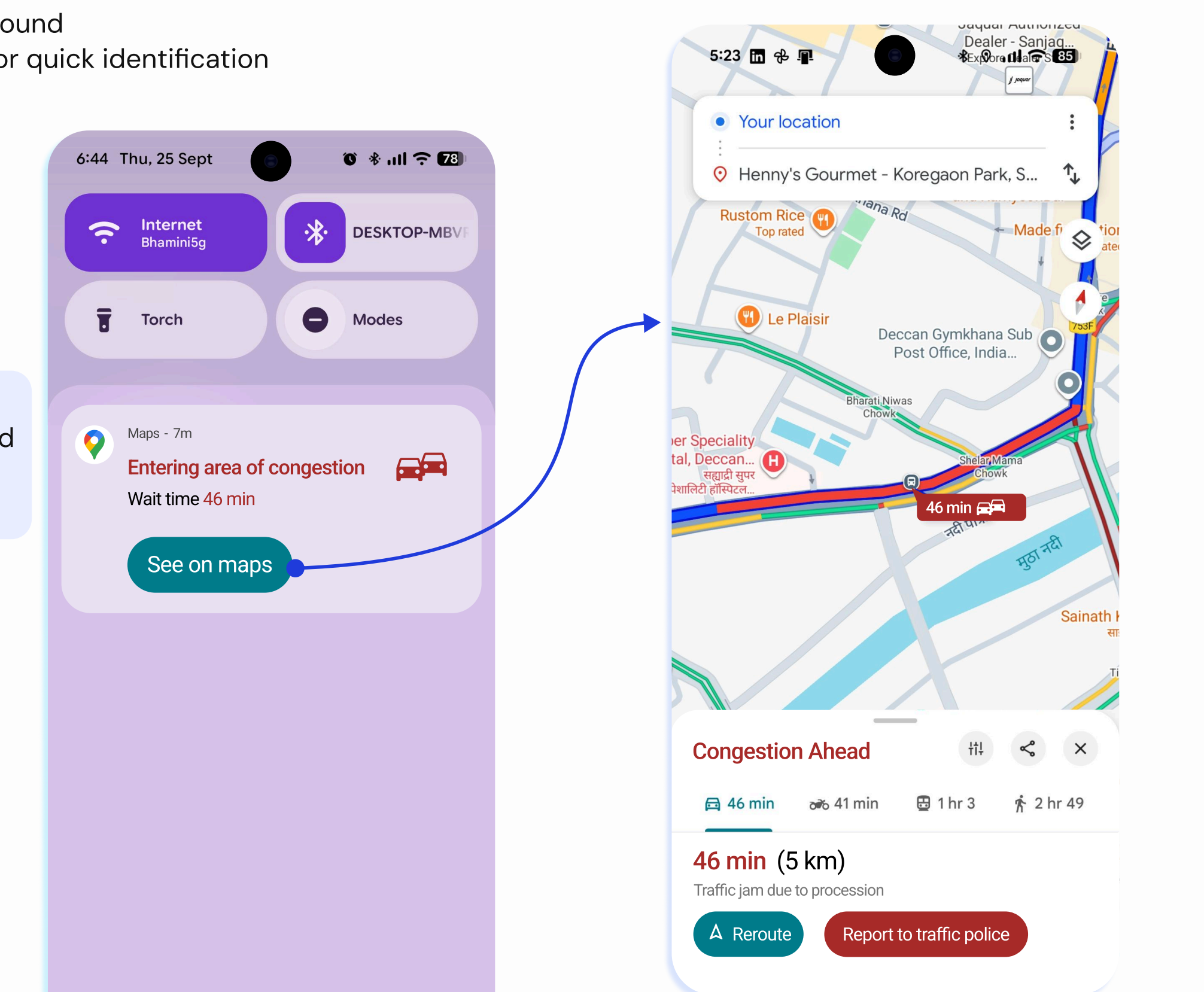
Flow 3 needs :
- Difference between travel time spend in congestion and normal travel time
- Option to avoid congestion
- Estimate time to leave to avoid the congestion

06 Solutions: UI Screens

1 Solution For "Entering Congestion" Users

For these people they need :
- Distinct notification sound
- Pop-up notification for quick identification

A pop-up sound that is instantly identifiable and clearly recognizable as related to traffic.


6:44 Thu, 25 Sept
Internet Bhamini5g
DESKTOP-M
Torch
Modes
Maps · 7m
See on maps
Your location
Henny's Gourmet - Koregaon Park, S...
46 min
Reroute
Report to traffic police

Entering area of congestion

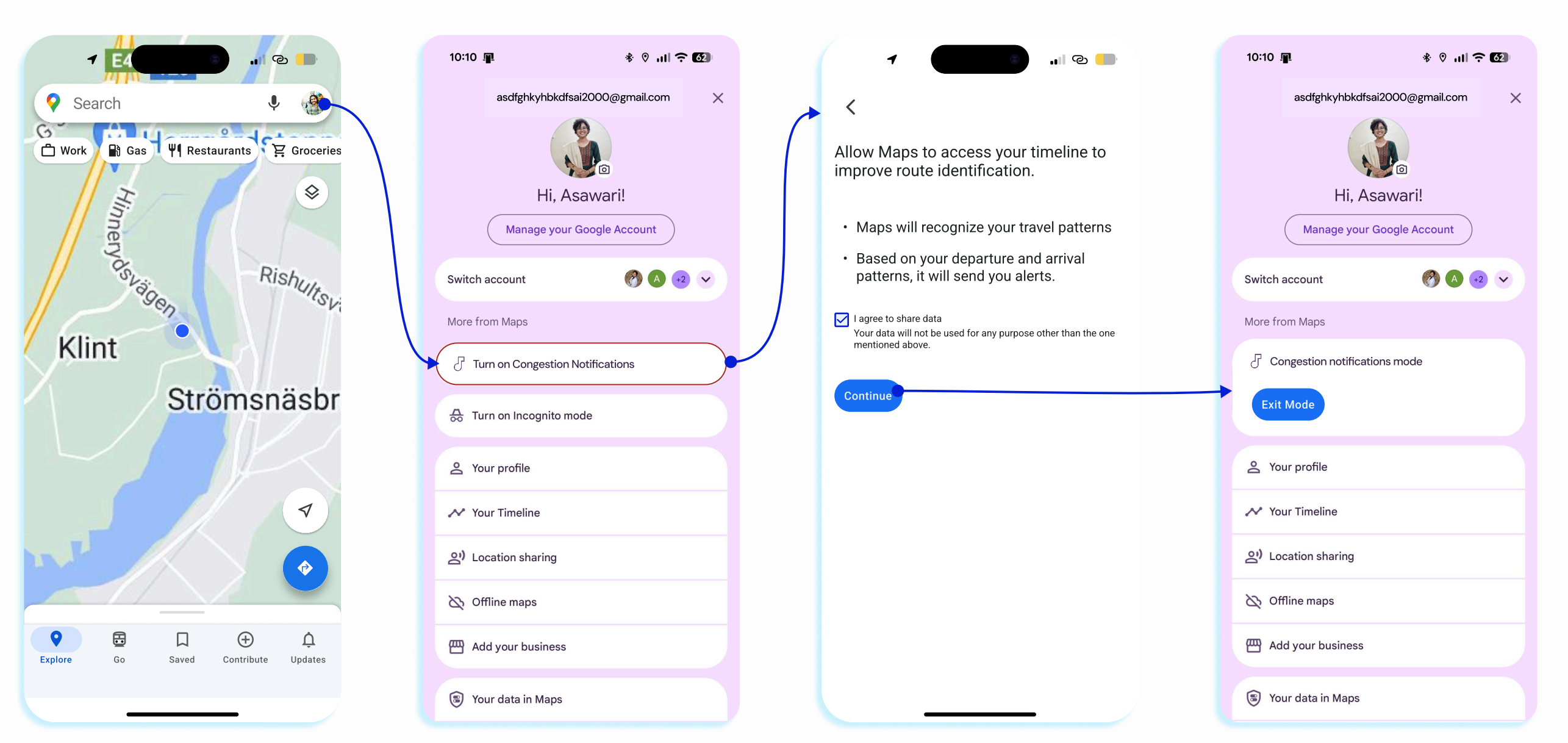
Wait time 46 min

Congestion Ahead

46 min (5 km)

Traffic jam due to procession

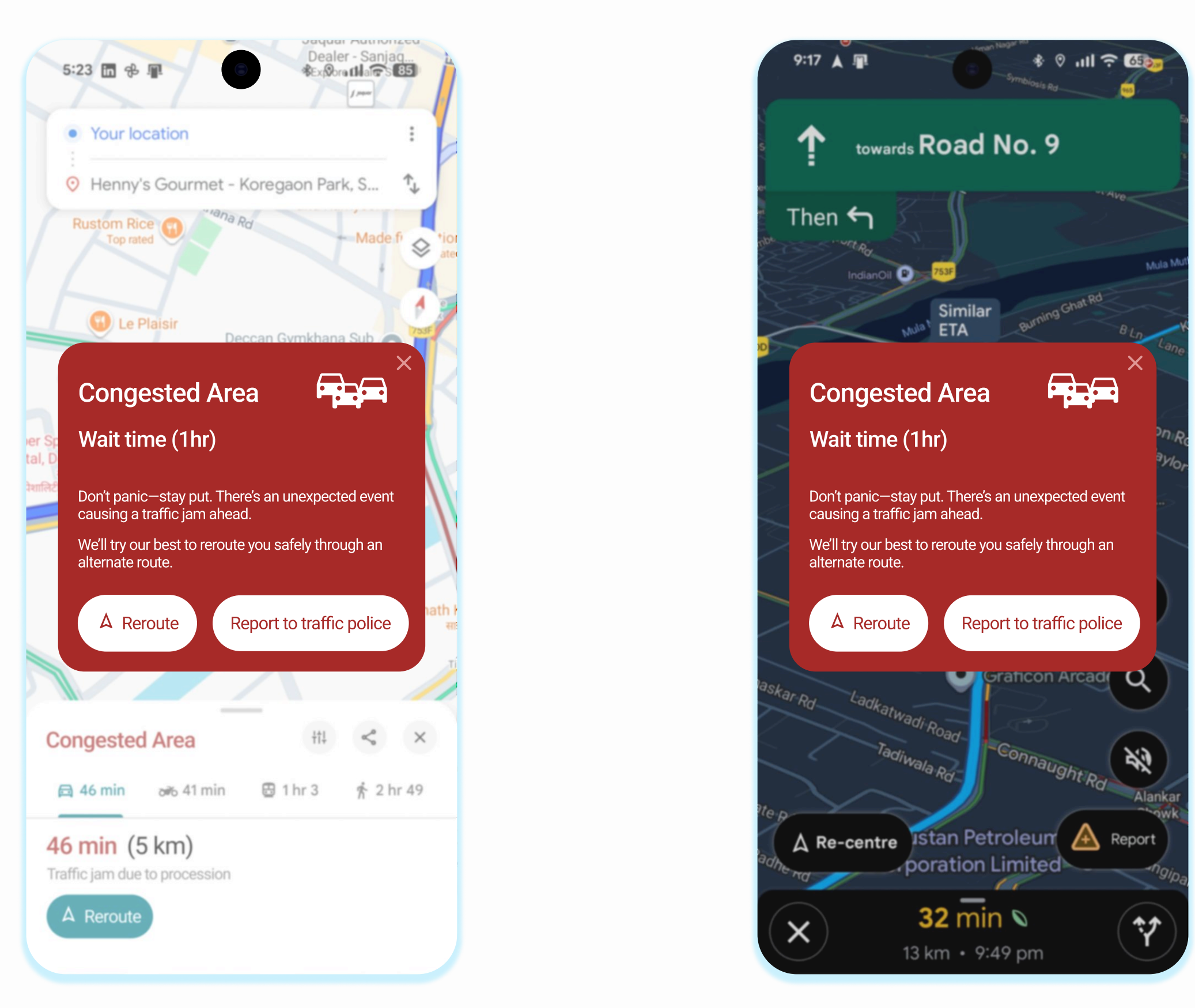
Steps to this utilize feature


Search
Klint
Strömsnäsbr
Hi, Asawari!
Manage your Google Account
Switch account
Turn on Congestion Notifications
Turn on incognito mode
Your profile
Your Timeline
Location sharing
Offline maps
Add your business
Your data in Maps
Continue
Congestion notifications mode
Exit Mode

Allow Maps to access your timeline to improve route identification.
- Maps will recognize your travel patterns
- Based on your departure and arrival patterns, it will send you alerts.

2 Solution For "Stuck in Traffic" Users

For users who are stuck in congestion they need :
- *Emotional* reassurance
- Estimated *waiting in congestion highlighted*
- Ability to *call traffic* authorities


Your location
Henny's Gourmet - Koregaon Park, S...
Reroute
Report to traffic police
46 min
41 min
1 hr 3
2 hr 49
towards Road No. 9
Then
Re-centre
Report
32 min
13 km · 9:49 pm

Congested Area

Wait time (1hr)

Don't panic—stay put. There's an unexpected event causing a traffic jam ahead.

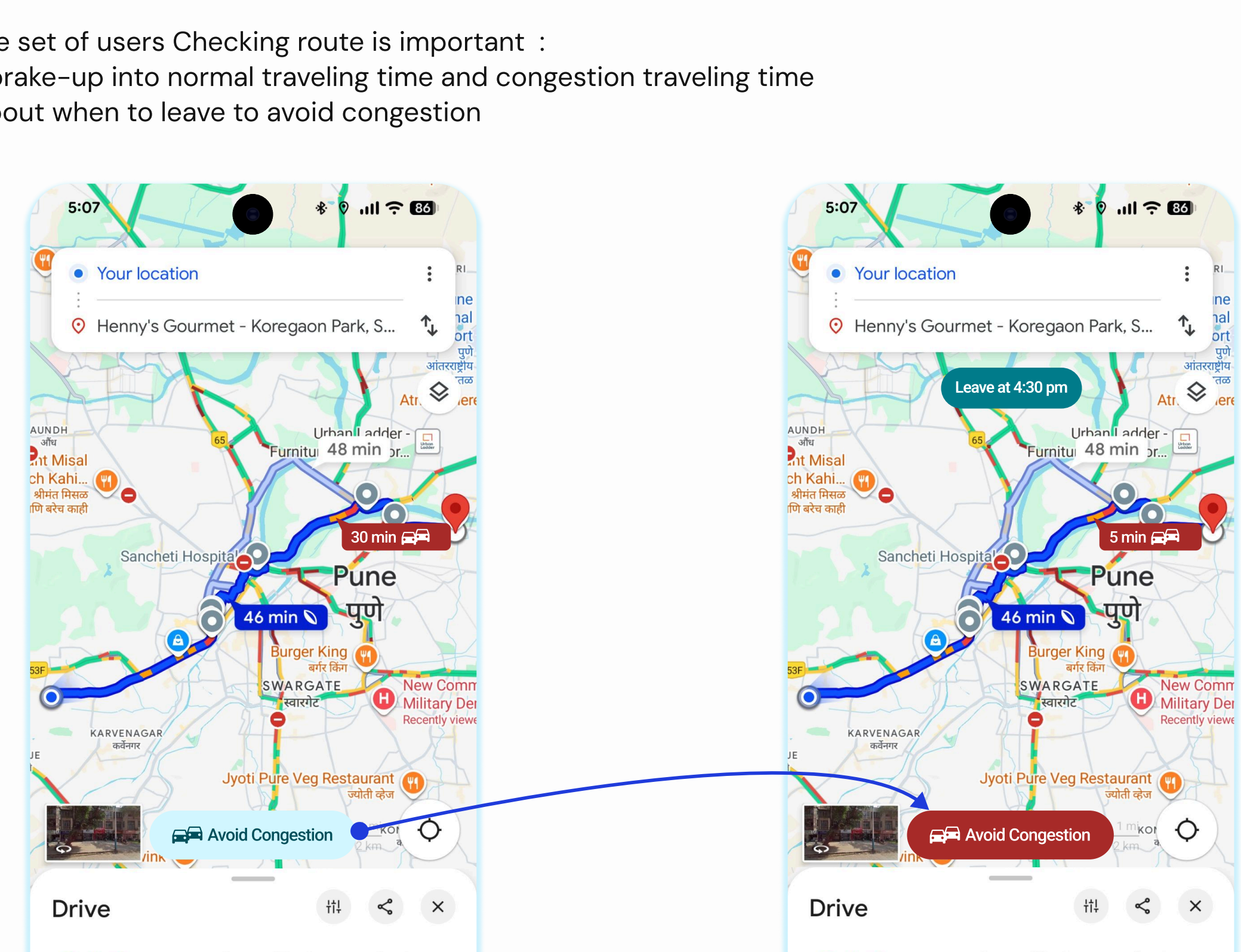
We'll try our best to reroute you safely through an alternate route.

46 min (5 km)

Traffic jam due to procession

3 Solution For "Checking Route" Users

For these set of users Checking route is important :
- ETA brake-up into normal traveling time and congestion traveling time
- ET about when to leave to avoid congestion


Your location
Henny's Gourmet - Koregaon Park, S...
30 min
46 min
48 min
5 min
4 min
Pune
Avoid Congestion
Leave at 4:30 pm
Start
Add stops
Share
Save

Drive

46 min (14 km) (6 min + 30 min)

Fastest route, despite the usual traffic

Saves petrol

Drive

46 min (14 km) (41 min + 5 min)

Leave at 4:30 pm to avoid congestion

Saves petrol

07 Conclusion

As society grows and evolves, it creates new problems—and old solutions often stop serving their original purpose. As designers, we must be sensitive enough to recognize these shifts and respond with thoughtful updates that feel like a natural extension of the existing system, not something abruptly new.

Just as a judicial system updates its laws when new cases challenge their relevance, it doesn't rewrite the entire constitution it refines and amends what already exists.

Similarly, in design, especially for products that are heavily used and deeply relied upon, we should adopt the same approach: evolve the system carefully, strengthen what works, and introduce changes that integrate seamlessly into users' everyday experiences.

08 References and Links

1. Research Papers
2. Analysis Report

FIN

Sniff— Sniff—